

Name: _____ Roll Number: _____

Question:	1	2	3	4	5	6	Total
Points:	0	5	5	5	7	4	26
Score:							

1. Every question perfectly solved below will add an additional 50% of the question's points to your total.

2. (5 points) **Parameter Estimation Error of the Minimum-Norm Estimator**

Consider the linear model $\underline{y} = A\underline{x}_* + \underline{\varepsilon}$, where $\underline{x}_* \in \mathbb{R}^d$ is the true parameter vector, $A \in \mathbb{R}^{n \times d}$ is a fixed design matrix, and $\underline{\varepsilon} \sim \mathcal{N}(\underline{0}, \sigma^2 I_n)$ is observation noise. Assume the *overparameterized* regime $n < d$, where A is a wide matrix with full row rank.

Let $A = U\Sigma V^\top$ be the thin SVD with singular values $\sigma_1 \geq \dots \geq \sigma_n > 0$ (so $U \in \mathbb{R}^{n \times n}$, $\Sigma \in \mathbb{R}^{n \times n}$, $V \in \mathbb{R}^{d \times n}$). As we derived in the assignment, the *minimum-norm* estimator of $\underline{x}_*$ from $\underline{y}$ is

$$\hat{\underline{x}}_{\text{mn}} = A^\top (AA^\top)^{-1} \underline{y} = V \Sigma^{-1} U^\top \underline{y}.$$

This is the unique solution to $\min_{\underline{x}} \|\underline{x}\|^2$ subject to $A\underline{x} = \underline{y}$.

Derive the parameter estimation error

$$(1/d) \mathbb{E}[\|\hat{\underline{x}}_{\text{mn}} - \underline{x}_*\|^2].$$

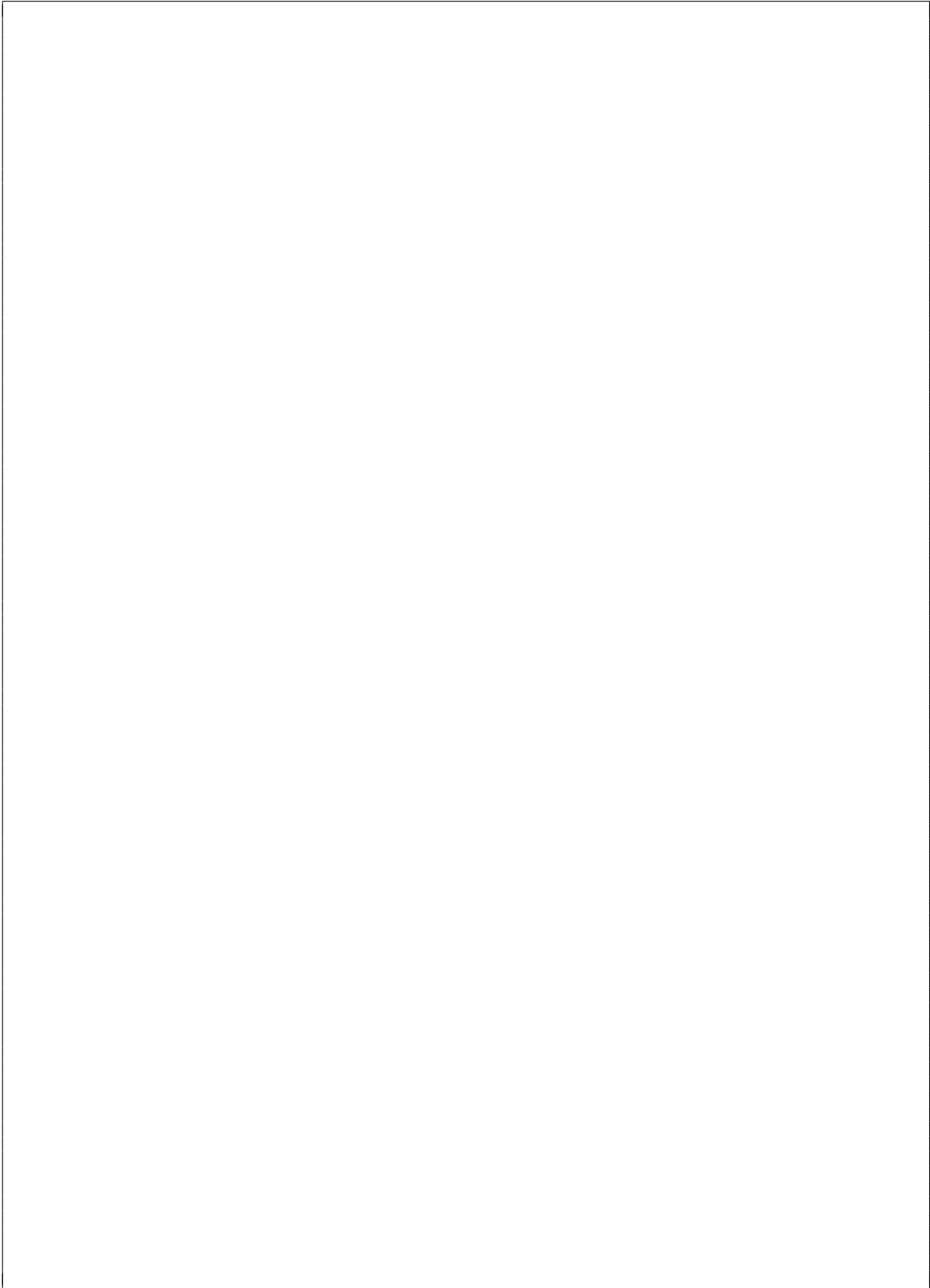
Express your answer in terms of $\underline{x}_*$, V , and the singular values $\{\sigma_i\}$.

Hints:

- Decompose $\underline{x}_*$ into orthogonal components as

$$\underline{x}_* = \underbrace{(I - VV^\top)\underline{x}_*}_{\underline{x}_0 \in \text{null}(A)} + \underbrace{VV^\top \underline{x}_*}_{\in \text{col}(V)}.$$

- If $\underline{z} \sim \mathcal{N}(\underline{\mu}, C)$, then $M\underline{z} \sim \mathcal{N}(M\underline{\mu}, MCM^\top)$.



3. (5 points) Gradient Descent Convergence: Identifying Step Sizes

Consider gradient descent on a 2-dimensional quadratic:

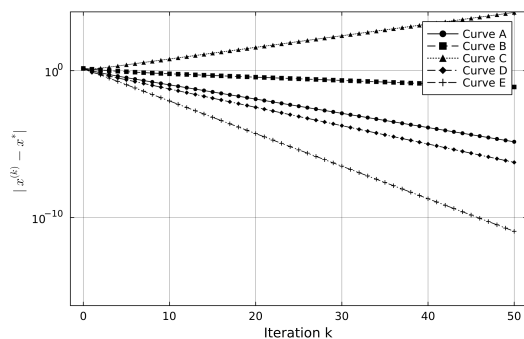
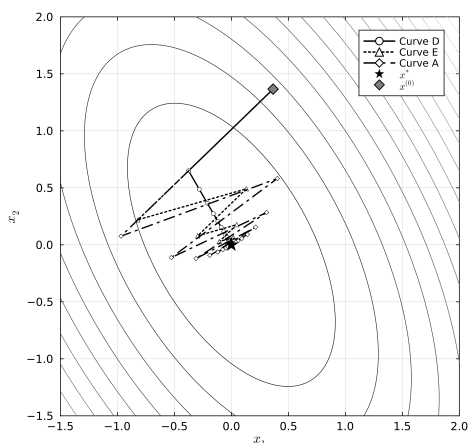
$$\min_{\underline{x} \in \mathbb{R}^2} f(\underline{x}) = \frac{1}{2} \underline{x}^\top H \underline{x} - \underline{b}^\top \underline{x},$$

where H is a 2×2 symmetric positive definite matrix with eigenvalues $\lambda_1 = 4$ and $\lambda_2 = 1$ (condition number $\kappa = 4$). The GD update is $\underline{x}^{(k+1)} = \underline{x}^{(k)} - \eta(H\underline{x}^{(k)} - \underline{b})$.

Recall from class that in the eigenbasis of H , the error in each direction satisfies

$$\alpha_i^{(k)} - \alpha_i^* = (1 - \eta\lambda_i)^k (\alpha_i^{(0)} - \alpha_i^*), \quad i = 1, 2.$$

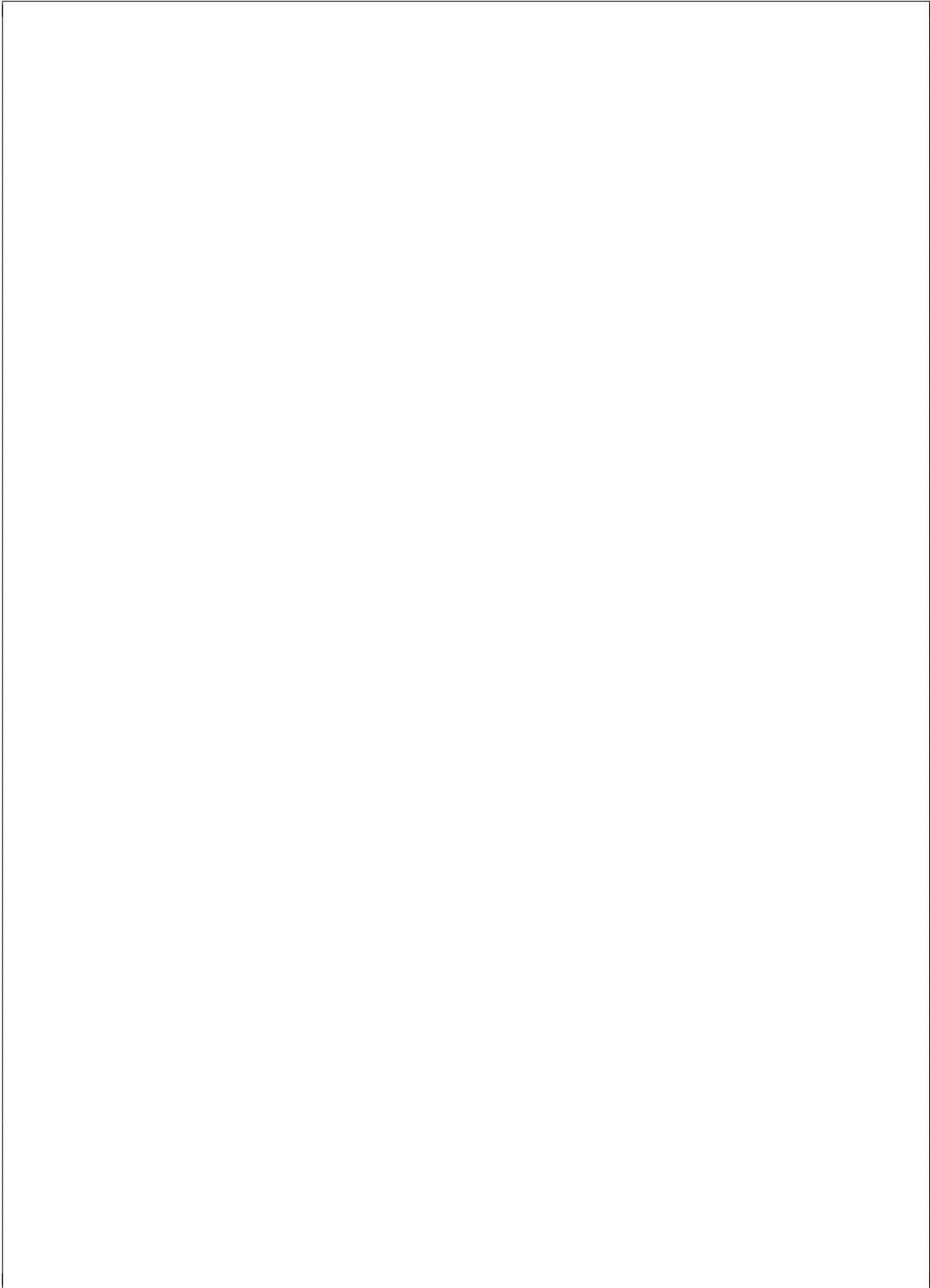
The plots below show the GD iterates for five different step sizes η , each starting from the same initial point $\underline{x}^{(0)}$, chosen so that $\underline{x}^{(0)} - \underline{x}^*$ has nonzero components along *both* eigenvectors of H . Plot 1 shows the 2D trajectories taken by the iterates (for three of the five step sizes) overlaid on the contours of f ; Plot 2 shows $\|\underline{x}^{(k)} - \underline{x}^*\|$ vs. iteration k on a log-scale y -axis for all five step sizes.



The five step sizes used are (in no particular order):

- (i) $\eta = 0.05$
- (ii) $\eta = 1/\lambda_1 = 0.25$
- (iii) $\eta = 2/(\lambda_1 + \lambda_2) = 0.4$
- (iv) $\eta = 0.45$
- (v) $\eta = 0.55$

Match each **Curve (A–E)** to the corresponding step size. For each match, briefly justify your answer by computing or reasoning about the contraction factors $(1 - \eta\lambda_1)$ and $(1 - \eta\lambda_2)$.

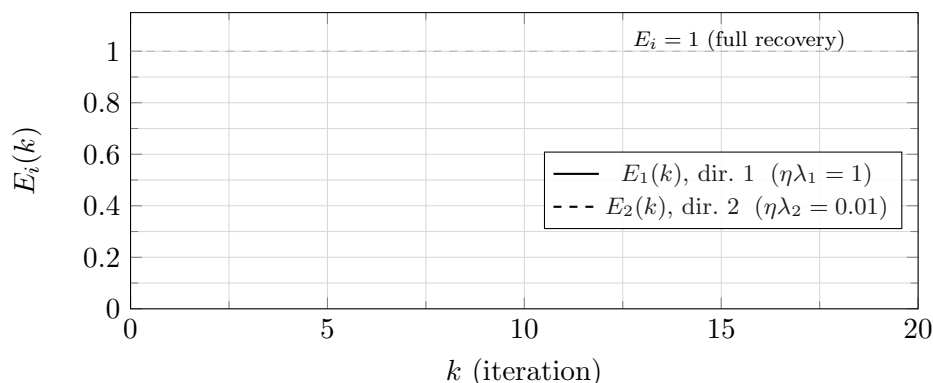


4. Spectral Bias: What Does GD Learn First?

Consider the linear model $\underline{y} = A\underline{x}_* + \underline{\varepsilon}$ with $A \in \mathbb{R}^{n \times 2}$, $\underline{x}_* \in \mathbb{R}^2$, and $\underline{\varepsilon} \sim \mathcal{N}(\underline{0}, \sigma^2 I_n)$. Let $H = \frac{1}{n} A^\top A$ have eigenvalues $\lambda_1 = 10$ and $\lambda_2 = 0.1$ with corresponding eigenvectors $\underline{v}_1, \underline{v}_2$. The true parameter $\underline{x}_*$ has equal energy in both eigen-directions: $\alpha_1^* = \underline{v}_1^\top \underline{x}_* = 1$ and $\alpha_2^* = \underline{v}_2^\top \underline{x}_* = 1$.

We run GD with step size $\eta = 1/\lambda_1 = 0.1$ starting from $\underline{x}^{(0)} = \underline{0}$.

- (a) (2 points) Let $\alpha_i^{(k)} = \underline{v}_i^\top \underline{x}^{(k)}$ denote the component of the k -th iterate along eigenvector $\underline{v}_i$. We let $E_i(k) = (\alpha_i^{(k)})^2 / (\alpha_i^*)^2$ for $i = 1, 2$ be the fraction of energy recovered after k iterations in direction i . Roughly sketch $E_i(k)$ for $i = 1, 2$ as function of k .



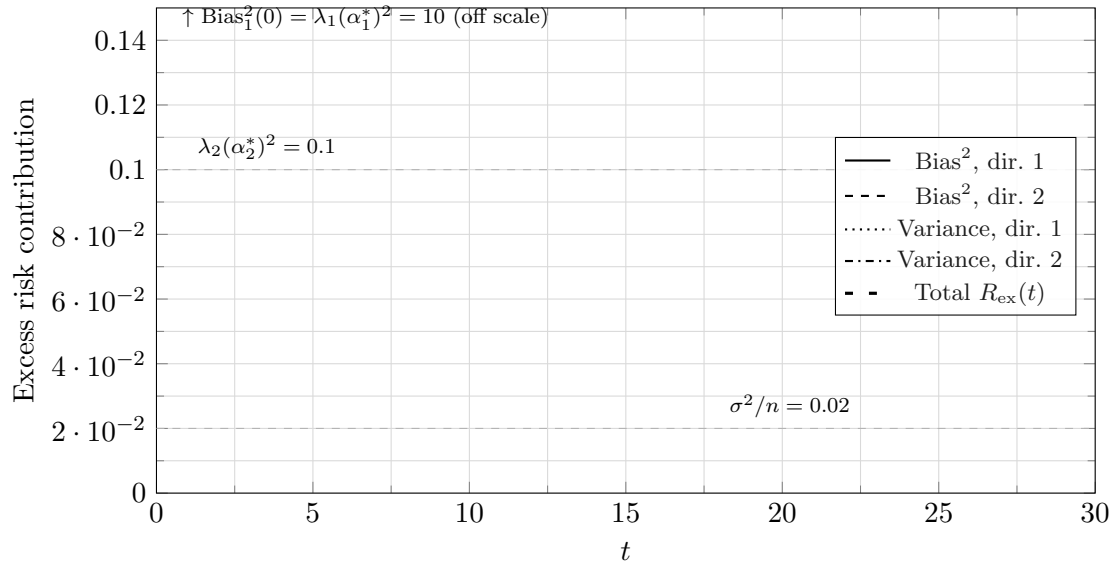
- (b) (2 points) As derived in class (Lecture 10.1), the excess risk of GD at continuous time t decomposes as

$$R_{\text{ex}}(t) = \underbrace{\sum_{i=1}^d \lambda_i e^{-2\eta\lambda_i t} (\alpha_i^*)^2}_{\text{Bias}^2(t)} + \underbrace{\sum_{i=1}^d \frac{(1 - e^{-\eta\lambda_i t})^2}{n}}_{\text{Variance}(t)} \sigma^2.$$

Note that both Bias^2 and Variance components have contributions from both eigen-directions.

Take $\sigma^2 = 1$ and $n = 50$. On the axes below, sketch the following four curves as functions of $t \geq 0$: Bias^2 and Variance contributions from directions 1 and 2. Mark the following on your plot:

- The value of each curve at $t = 0$ and its limiting value as $t \rightarrow \infty$.
- The approximate time at which direction 1's bias becomes negligible.
- The total excess risk $R_{\text{ex}}(t)$ (sum of all four curves). Indicate approximately where it is minimized.



- (c) (1 point) Recall that the Hessian H has the interpretation of the empirical correlation matrix of the data. The eigen-directions with large eigenvalues correspond to high-variance directions (important) in the data, while those with small eigenvalues correspond to low-variance (unimportant/noisy) directions.

Based on your observations, can we give an explanation on why early stopping acts as regularization? Which components of the signal does it effectively “discard,” and why is this beneficial when there is noise?

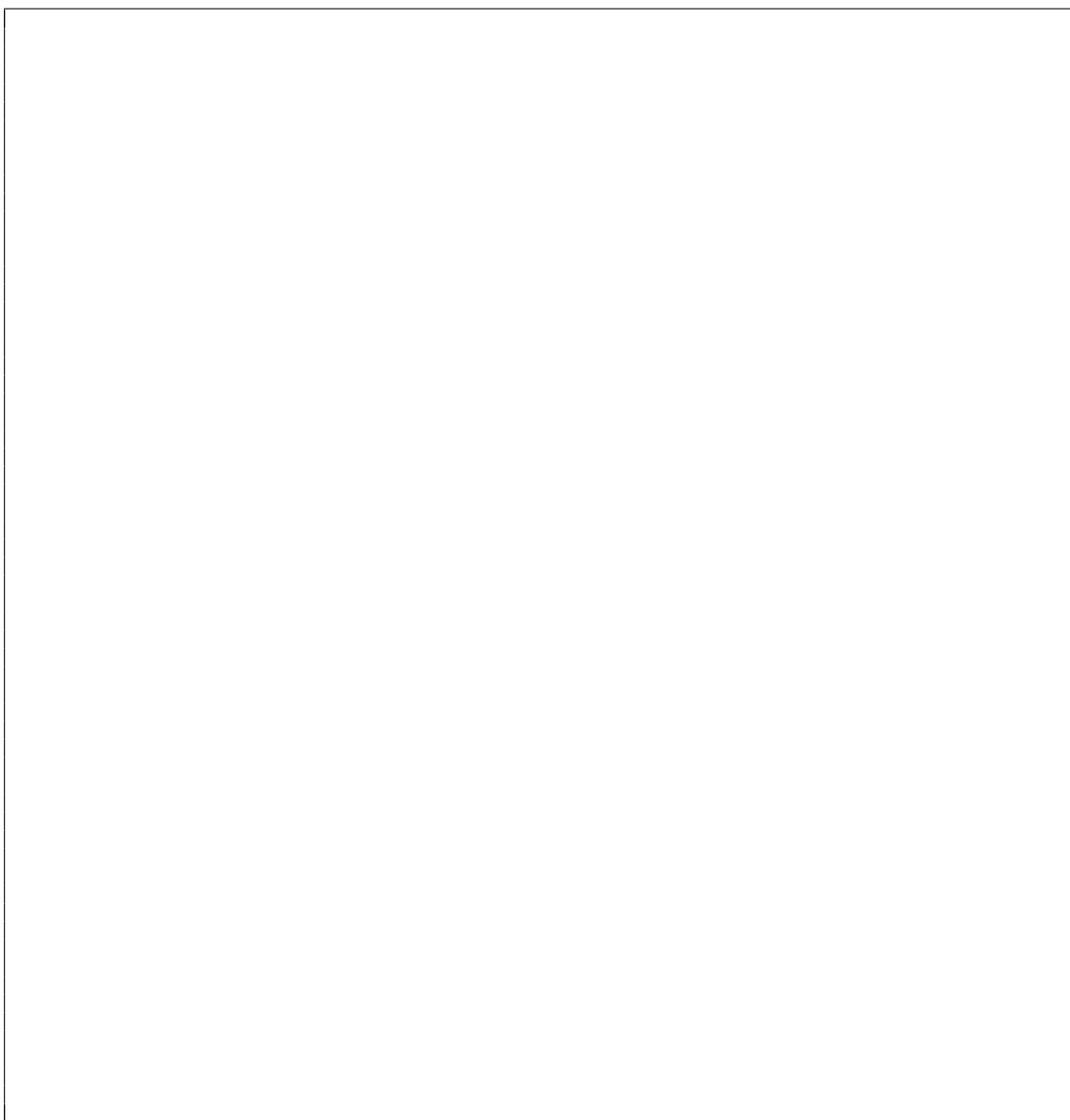
5. Co-coercivity and an Alternative GD Convergence Proof

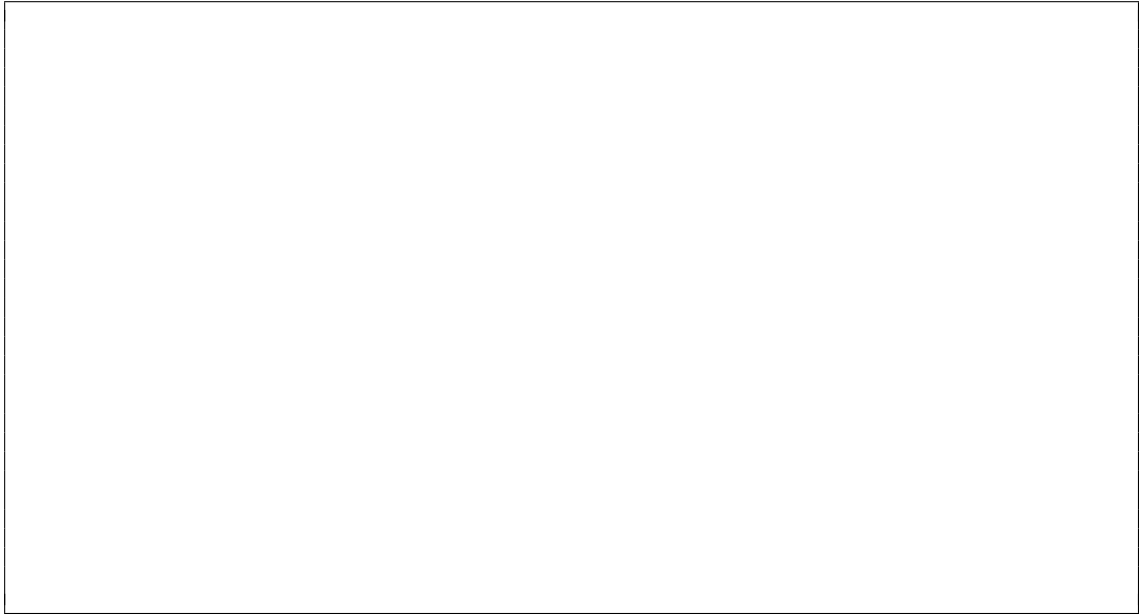
Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be closed, convex, and L -smooth, with minimizer $\underline{x}^*$.

You may use the following facts from class without proof:

- (i) If f is closed, convex, and L -smooth, then f^* is $\frac{1}{L}$ -strongly convex.
- (ii) If g is μ -strongly convex, then $(\nabla g(\underline{u}) - \nabla g(\underline{v}))^\top (\underline{u} - \underline{v}) \geq \mu \|\underline{u} - \underline{v}\|^2$ (gradient monotonicity).
- (iii) For closed, convex, differentiable f : $\nabla f^*(\nabla f(\underline{x})) = \underline{x}$.
- (a) (2 points) Prove the *co-coercivity* inequality:

$$(\nabla f(\underline{x}) - \nabla f(\underline{y}))^\top (\underline{x} - \underline{y}) \geq \frac{1}{L} \|\nabla f(\underline{x}) - \nabla f(\underline{y})\|^2 \quad \forall \underline{x}, \underline{y}.$$



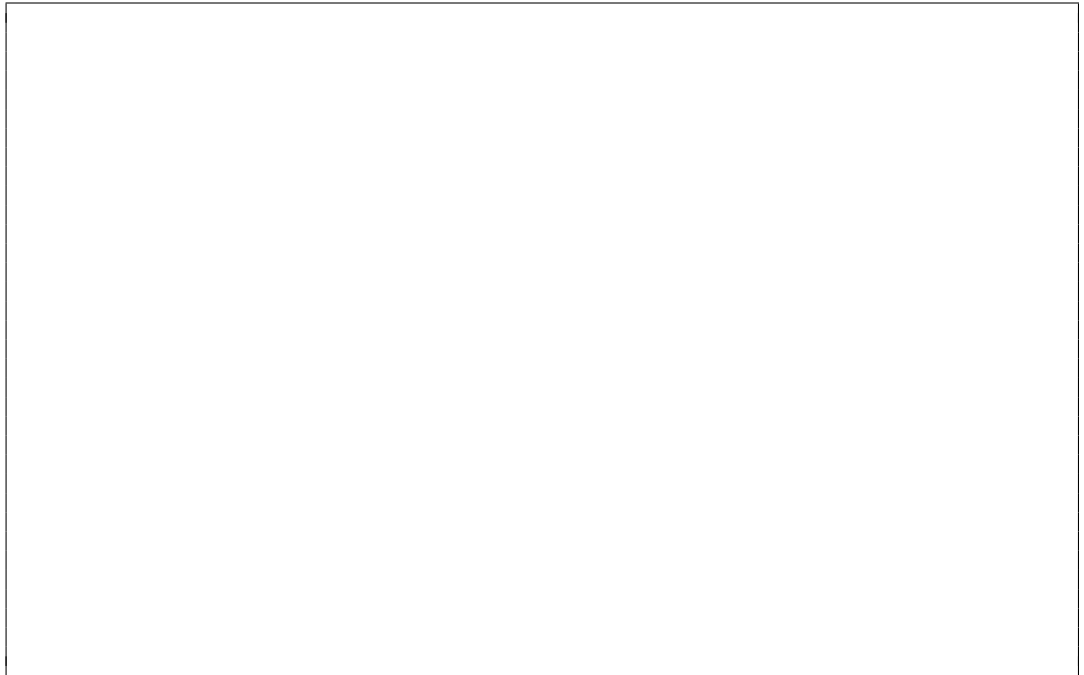


- (b) Consider GD with step size $\eta = 1/L$: $\underline{x}^{(k)} = \underline{x}^{(k-1)} - \frac{1}{L} \nabla f(\underline{x}^{(k-1)})$. The following is an alternative proof (to the one given in class) of convergence of GD for L-smooth functions. We will prove that

$$f(\underline{x}^{(T)}) - f(\underline{x}^*) \leq \frac{L}{T} \left\| \underline{x}^{(0)} - \underline{x}^* \right\|^2.$$

Justify each of the three steps below.

- (a) (2 points) **Step 1.** $\left\| \underline{x}^{(k)} - \underline{x}^* \right\|^2 \leq \left\| \underline{x}^{(k-1)} - \underline{x}^* \right\|^2 - \frac{1}{L} \nabla f(\underline{x}^{(k-1)})^\top (\underline{x}^{(k-1)} - \underline{x}^*)$.
Hint: Expand the LHS using the GD update and apply co-coercivity from part (a).



(b) (2 points) **Step 2.** $f(\underline{x}^{(k)}) - f(\underline{x}^*) \leq L \left(\|\underline{x}^{(k-1)} - \underline{x}^*\|^2 - \|\underline{x}^{(k)} - \underline{x}^*\|^2 \right).$

(c) (1 point) **Step 3.** $f(\underline{x}^{(T)}) - f(\underline{x}^*) \leq \frac{L}{T} \|\underline{x}^{(0)} - \underline{x}^*\|^2.$

6. SGD Convergence with Alternative Learning Rate Schedules

Recall from Lecture 10 that for SGD on least squares with step size schedule η_t , the continuous-time relaxation of the expected squared error $E(t) = \mathbb{E} \|\underline{e}(t)\|^2$ satisfies

$$\frac{dE}{dt} = -2\eta(t) \lambda_{\min} E(t) + \eta(t)^2 B,$$

where $\lambda_{\min} > 0$ is the smallest eigenvalue of $H = \frac{1}{n} A^\top A$ and $B > 0$ is a constant bounding the stochastic gradient noise. In class, we solved this for $\eta(t) = c/t$ (with $2c\lambda_{\min} > 1$) and obtained $E(t) = O(1/t)$.

Now analyze two other schedules.

- (a) (2 points) $\eta(t) = c/t^2$. Write down the ODE for this schedule. As a hint, first solve the noiseless case $B = 0$, noting that the general solution for $B > 0$ includes the homogeneous ($B = 0$) solution as one of its terms. Does $E(t) \rightarrow 0$ as $t \rightarrow \infty$?

- (b) (2 points) $\eta(t) = c/\sqrt{t}$. With noise ($B > 0$), Find the asymptotic behavior of $E(t)$ by guessing $E(t) = D/t^p$ (for constants $D > 0$ and $p > 0$), substituting into the ODE, and comparing powers of t^{-1} on both sides. What convergence rate do you obtain? Is it faster or slower than the $O(1/t)$ rate of $\eta(t) = c/t$?